

Course Project Classification

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ISQA 8720

1. COURSE PROJECT DESCRIPTION AND BUSINESS UNDERSTANDING

Nebraska requires that all public school students in the third year of their cohort take the ACT college entrance exam [1]. To prepare students for this standardized test, Bennington Public Schools (BPS) requires that its first- and second-year cohorts also take the Pre-ACT [2]. An ACT score at or above 19 is considered by BPS to be a successful ACT score. Identifying students near this score would enable BPS to take action, potentially guiding students to stay at or above this critical score. Hence, the goal of the project is to predict which students fall within the critical range (ACT score of 18-20) for their 11th-grade spring semester ACT test. This project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, a standard framework for data science projects that moves through six phases: business understanding, data understanding, data preparation, modeling, evaluation, and deployment. Each section of this report corresponds to one of these phases.

Various information about the BPS students has been collected, including previous test scores, demographics, and more. The collection of this information or features can be used to predict which students fall within the defined 18-20 ACT score range. A statistical machine learning model approach can help to discern which features are more or less useful than others. These useful features can then be used to train a model to predict which of the students fall within the critical range.

The outcome of the project would identify a subset of students predicted to receive an ACT score within the range of 18-20. This would aid in monitoring students' performance as they progress through high school until they take the ACT in the spring semester of their junior year, and could provide assistance if a student struggles in a specific area before then. Additionally, the classification of these students will help to identify influential factors that make it more or less likely a student will fall within this critical range. By identifying these factors, school staff can try to curb or increase them overall and monitor them for each student.

2. DATA UNDERSTANDING

We have a wealth of data, including demographic, curricular, extracurricular, and behavioral information, as well as Pre-ACT and ACT scores for BPS high school students from the past 5 years. This data was collected internally by BPS and handed over to us. While mostly complete, the data has some missing values that need to be addressed. This data could be useful for classifying students with ACT scores in the 18-20 range.

One important characteristic of the dataset is class imbalance in the target variable. Of the students in the dataset, approximately 22% fall within the critical ACT range of 18-20, while the remaining 78% do not. This imbalance is significant because machine learning models tend to default toward predicting the majority class, meaning a naive model would simply classify most students as ‘No’ and thereby achieve high accuracy while being completely useless at actually identifying at-risk students. This class imbalance directly informed our modeling strategy, which is discussed further in Section 4.

The data is structured in a large CSV file. This contains 50 fields or columns for each observation. There are about 5,400 observations, which represent a semester for a particular student. There are a multitude of data types across fields, such as numerical, categorical, date, string, and boolean. For the field of interest, our target variable, ACT_Critical_Range, is a boolean.

Various fields can be clustered into a few categories. There are the standardized test scores of the ACT, PreACT, and other scores from the MAP and NSCAS tests. Additionally, there is demographic information like ethnicity, gender, and more. Also, there is curricular information on GPA and credits. The data also includes behavioral information, such as suspensions, nurse visits, and tardies.

Descriptive Statistics:

Data Summary	
Name	values
Number of Rows	5378
Number of Columns	50

Column Type Frequency	
Character	15
Logical	5
Numeric	30

Logical & Character Fields

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
SchoolYear	0	1	9	9	0	6	0
Semester	0	1	2	2	0	2	0
Gender	0	1	4	6	0	2	0
Ethnicity	0	1	1	1	0	7	0

SPED	0	1	2	3	0	2	0
X504	0	1	2	3	0	2	0
HAL	0	1	2	3	0	2	0
Free_and_Reduced_Lunch	0	1	2	3	0	2	0
Summer_School	0	1	2	3	0	2	0
Athletic_Participation	0	1	2	3	0	2	0
Activities_Participation	0	1	2	3	0	2	0
NSCAS_ELA_Proficiency _8th_Grade	0	1	0	24	2897	4	0
NSCAS_Math_Proficiency _8th_Grade	0	1	0	24	2905	4	0
NSCAS_Science_Proficien cy_8th_Grade	0	1	0	17	4120	5	0
ACT_Critical_Range	0	1	2	3	0	2	0

PreACT_Composite_8_9, PreACT_English_8_9, PreACT_Reading_8_9, PreACT_Science_8_9,
PreACT_Math_8_9 are all NaNs

Numeric Fields

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75
StudentID	0	1.0000000	1.540912e+03	309.2597393	1007.000	1282.00	1524.000	1790.00
Grd	0	1.0000000	1.056768e+01	1.0685856	8.000	10.00	11.000	11.00
GraduationCohort	0	1.0000000	2.024566e+03	1.0880064	2022.000	2024.00	2025.000	2026.00
Behavior_Referrals	0	1.0000000	7.437709e-03	0.1868379	0.000	0.00	0.000	0.00
DaysAbsent	0	1.0000000	4.888180e+00	5.1854958	0.000	1.75	3.630	6.50
Number_Tardies	0	1.0000000	2.437337e+00	5.5761151	0.000	0.00	0.000	2.00
Nurse_Visits	0	1.0000000	2.361473e-02	0.1839784	0.000	0.00	0.000	0.00
Number_OSS_Occ urrences	0	1.0000000	2.138341e-02	0.1650857	0.000	0.00	0.000	0.00

Number_ISS_Occurrences	0	1.0000000	2.101153e-02	0.1640033	0.000	0.00	0.000	0.00
Number_OSS_Days	0	1.0000000	8.264597e-02	0.8280626	0.000	0.00	0.000	0.00
Number_ISS_Days	0	1.0000000	3.188918e-02	0.2700405	0.000	0.00	0.000	0.00
Last_Reading_MAP_Percentile	171	0.9682038	6.234185e+01	21.5600436	4.000	49.00	66.000	79.00
Last_Math_MAP_Percentile	104	0.9806620	7.071521e+01	21.8846962	3.000	57.00	75.000	88.00
PreACT_Composite	2694	0.4990703	1.974590e+01	4.7285899	11.000	16.00	20.000	23.00
PreACT_English	2694	0.4990703	1.718666e+01	5.3939642	5.000	14.00	17.000	21.00
PreACT_Reading	2694	0.4990703	2.171088e+01	6.4224971	5.000	17.00	21.000	26.00
PreACT_Math	2694	0.4990703	1.964083e+01	4.6990600	13.000	16.00	18.000	23.00
PreACT_Science	2694	0.4990703	1.993182e+01	5.0247439	9.000	17.00	20.000	23.00
Completed_Credit_Count	0	1.0000000	6.452399e+00	1.3145948	0.000	6.00	7.000	7.00
Num_Credits_Completed_English	0	1.0000000	1.102640e+00	0.3974399	0.000	1.00	1.000	1.00
Num_Credits_Completed_Math	0	1.0000000	9.466344e-01	0.4076824	0.000	1.00	1.000	1.00
Num_Credits_Completed_Science	0	1.0000000	9.053552e-01	0.4956245	0.000	1.00	1.000	1.00
GPA	0	1.0000000	3.231423e+00	0.7240034	0.646	2.75	3.346	3.88
ACT_Composite	3290	0.3882484	2.151389e+01	5.3183105	11.000	17.00	21.000	25.00

ACT_English	3290	0.3882484	2.080316e+01	6.1684585	8.000	16.00	20.000	24.00
ACT_Reading	3290	0.3882484	2.204693e+01	6.4097210	8.000	17.00	21.000	26.00
ACT_Math	3290	0.3882484	2.110920e+01	5.4148947	10.000	17.00	20.000	25.00
ACT_Science	3290	0.3882484	2.160536e+01	5.3859575	7.000	18.00	21.000	24.00
ACT_Composite_Junior_Spring	0	1.0000000	2.143864e+01	5.3035485	11.000	17.00	21.000	25.00
ACT_Composite_Max	0	1.0000000	2.156471e+01	5.3384256	11.000	18.00	21.000	25.00

Several patterns are worth noting in the descriptive statistics. The average GPA across all observations is 3.23, though the range of 0.65 to 3.88 indicates considerable variation among students. The median PreACT Composite score is 20, just at the upper boundary of the critical range, suggesting that a meaningful portion of students are clustered around the 18-20 threshold. Behavioral fields such as suspensions, nurse visits, and behavior referrals are very sparse, which limits their individual predictive power but may still have useful predictive value. The PreACT fields are missing for roughly 50% of observations, which is the most significant data quality concern in the dataset and is addressed through mean imputation in section 3.

All logical & character fields are filled in, except NSCAS_ELA_Proficiency_8th_Grade, NSCAS_Math_Proficiency_8th_Grade, and NSCAS_Science_Proficiency_8th_Grade, which have quite a few empty values. Our target variable, ACT_Critical_Range, is a part of the logical fields.

For numerical fields, there are quite a few fields that have missing values, like Last_Reading_MAP_Percentile, Last_Math_MAP_Percentile, PreACT_Composite, PreACT_English, PreACT_Reading, PreACT_Math, PreACT_Science, ACT_Composite, ACT_English, ACT_Reading, ACT_Math, and ACT_Science. The PreACT fields having this many empty values are concerning, but may be mitigated by structuring the data differently, since the PreACT is repeated for every observation. Some behavioral fields are very sparse, possibly because the data is ordered by semester; this may be mitigated by restructuring the data and applying transformations.

Some fields that we expected to be dynamic are actually static. For example, a student's GPA is expected to change each semester, but it is the same across observations of that student. Another important field that is unexpectedly static is the PreACT field, which is either NA or a numeric score and does not change for a given student across semesters. No other external data was included in the data set.

3. DATA PREPARATION

We dropped fields ACT_Composite, ACT_English, ACT_Reading, ACT_Math, ACT_Science, ACT_Composite_Junior_Spring, and ACT_Composite_Max as they directly indicate the ACT_Critical_Range target variable. These fields are target variables for goal #1, thus not relevant to us, as we are attempting goal #2.

We dropped the fields SchoolYear, Semester, Grd, and GraduationCohort because the dataset is structured by studentID. The idea being that this is a snapshot in time just before the student's spring semester of junior year. We dropped fields that have no values: PreACT_Composite_8_9, PreACT_English_8_9, PreACT_Reading_8_9, PreACT_Science_8_9, PreACT_Math_8_9, NSCAS_ELA_Proficiency_8th_Grade, NSCAS_Math_Proficiency_8th_Grade, and NSCAS_Science_Proficiency_8th_Grade.

Fields that are directly imported from the original set have only one observation, which is usually repeated for every observation in time for that student. ie) The same GPA is repeated for every semester/observation for a given student, or the same Ethnicity is repeated for every semester/observation for a given student (these imported fields do not change for a student in the original data, so they can simply be pulled from one observation of a student; they are static). There are no extreme outliers in the original data or the new data set we created.

We have discussed the benefits of creating derived variables for this project, as some values or groups of values will benefit from composite derived variables. We created several new variables derived from the original data set. Most of the derived variables are means. For example, Athletic_Participation is a boolean in the original set, and in the transformed dataset, we take the average number of "yes" responses. Another type of data we averaged is the Days_Absent field; since it is numeric, we simply take its mean for the new data set. Averages were the primary method for transforming fields, as we believed this made the most sense for those fields. Below, we outline all the data fields included in this project, indicating the type of transformation applied to each and its data type.

Data Fields Template								
Data Transformation	StudentID	ACT_Critical_Range	Gender	Ethnicity	If SPED Is "yes" once	If X504 is "yes" once	If HAL is "yes" once	GPA
Data Type	Categorical	Boolean	Boolean	Categorical	Boolean	Boolean	Boolean	Numerical

Data Fields Template continued				
Data Transformation	Mean number of “yes” for Free_and_Reduced_Lunch up to grade 12 S1	Mean number of “yes” for Summer_School up to grade 12 S1	Mean number of “yes” for Athletic_Participation up to grade 12 S1	Mean number of “yes” for Activites_Participation up to grade 12 S1
Data Type	Numerical	Numerical	Numerical	Numerical

Data Fields Template continued				
Data Transformation	Mean of Behavior Referrals up to grade 12 S1	Days_Absent up to grade 12 S1 (repeated feature for every semester)	Number_Tardies up to grade 12 S1 (repeated feature for every semester)	Mean of Nurse_Visits up to grade 12 S1
Data Type	Numerical	Numerical	Numerical	Numerical

Data Fields Template continued				
Data Transformation	Mean of Number_OSS_Occurences up to grade 12 S1	Mean of Number_ISS_Occurrences up to grade 12 S1	Mean of Number_OSS_Days up to grade 12 S1	Mean of Number_ISS_Days up to grade 12 S1
Data Type	Numerical	Numerical	Numerical	Numerical

Data Fields Template continued				
Data Transformation	Last_Reading_MAP_Percentile	Last_Math_MAP_Percentile	NSCAS_ELA_Proficiency_8th_Grade	NSCAS_Math_Proficiency_8th_Grade
Data Type	Numerical	Numerical	Categorical	Categorical

Data Fields Template continued				
Data Transformation	NSCAS_Science_Proficiency_8th_Grade	Mean of Completed_Credit_Count up to grade 12 S1	Mean of Num_Credits_Completed_English up to grade 12 S1	Mean of Num_Credits_Completed_Math up to grade 12 S1
Data Type	Categorical	Numerical	Numerical	Numerical

Data Fields Template continued				
Data Transformation	Mean of Num_Credits_Completed_Science up to grade 12 S1	PreACT_Composite (Static but with N/A values) For those students without this field, we will impute the mean	PreACT_English (Static but with N/A values) For those students without this field, we will impute the mean	PreACT_Reading (Static but with N/A values) For those students without this field, we will impute the mean
Data Type	Numerical	Numerical	Numerical	Numerical

Data Fields Template continued				
Data Transformation	PreACT_Math (Static but with N/A values) For those students without this field, we will impute the mean	PreACT_Science (Static but with N/A values) For those students without this field, we will impute the mean		
Data Type	Numerical	Numerical		

***BOLD** indicates static values

4. DATA MODELING, MODEL BUILDING RESULTS

For classifying students into the ACT critical range, Random Forest (RF) was first used. This was to gain insight into the importance of features and to guide us in performing data preparation differently based on the initial results, such as imputing features differently or removing certain features entirely. Another model we used was XGBoost (XGB), as it is often more accurate than random forests and handles imbalanced data better, which is needed here. In addition, we used a pruned decision tree, as this would allow for greater interpretability of how students end up in this critical range. Finally, we used a random classifier as a baseline for comparison against all these models.

To evaluate our models, we used a technique called k-fold cross-validation (CV). This works by splitting the training data into five equal groups, or “folds”. The model is trained on four of the five folds and tested on the remaining one; this process repeats five times, so that each fold is used as the test set once. We repeated this entire process three times with different random splits to get a more stable estimate of performance. This approach helps ensure that our results are not just a product of one lucky or unlucky data split.

A regular grid was used to tune the hyperparameters. For RF, these were mtry (number of predictors randomly sampled at each split), trees (number of trees), and min_n (minimum number of observations in a terminal node), with 5 levels each (5³ hyperparameter combinations). For XGB, these were trees (number of trees), min_n (minimum observations in a terminal node), tree depth, and the learning rate. The same cross-validation was used to evaluate the best-fit model on the training and validation sets, and then the resulting model was applied to the test set. For the decision tree, the cost complexity parameter was tuned to maximize ROC AUC.

The primary metric used to tune and compare the models was the Receiver Operating Characteristic Area Under the Curve (ROC AUC). This metric essentially measures how well a model can distinguish between the two classes, that is, in our case, how well the model is able to characterize the students in the

critical ACT range versus those outside of the range. A score of 1.0 represents a perfect model, while a score of 0.5 means the model performs no better than just randomly guessing. We also closely monitored the variables' sensitivity and specificity. Sensitivity measures how often the model correctly identifies students who actually fall within the critical range; that is, low sensitivity means many potentially at-risk students are missed or misclassified. Specificity, on the other hand, measures how often the model correctly identifies students who do not fall in the critical range.

4.1 RANDOM FOREST

Random Forest is an ensemble machine learning method that builds a large number of individual decision trees, each trained on a random subset of the data and features, and combines their predictions via majority voting. This approach reduces the risk of overfitting that a singular decision tree would have, and tends to produce robust, accurate predictions even on complex datasets. Prior to creating the random forest, we imputed/edited the variables (seen in the Data Preparation Section). We sought to understand our model by examining which predictors we were feeding into it. So we first created the RF model optimizing over AUC. To understand which variables were the most useful for prediction, we examined the feature importance. Feature importance tells us how much each variable contributed to the model's decisions. We used the Gini Index to measure this; it quantifies how much each variable reduces uncertainty at each split. Effectively, in practice, variables that consistently lead to cleaner, more decisive splits receive higher importance scores.

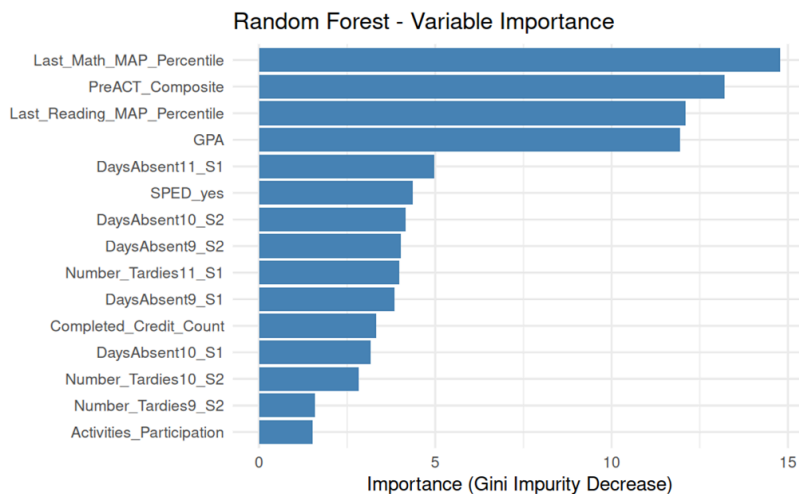


Figure 1. Feature importance of the top 15 features within the RF model of the first run. Utilizing the Gini Index for determining feature importance from this, we dropped any values below Num_Credits_Completed_English.

After selecting predictors based on feature importance, we repeated the RF procedure in a second run using the selected important features.

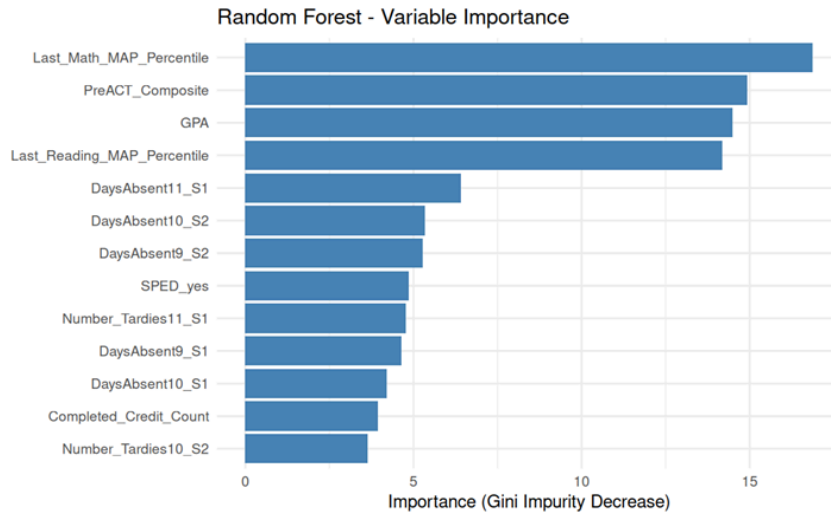


Figure 2. Feature importance of the features within the RF model of the second run. This Utilizes the Gini Index for determining feature importance.

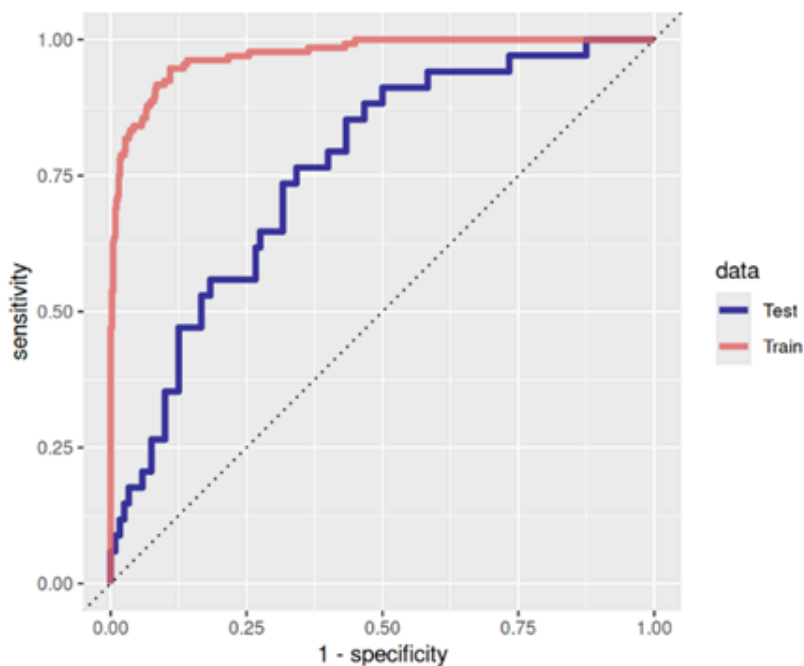


Figure 3. ROC AUC plot of the second RF run.

After the second run, we performed a third RF run to address the class imbalance in the target variable. There are marginally more “No” than “Yes” in the ACT critical range feature column. To address this, we synthetically imputed additional “Yes” observations for the ACT critical range target variable, ensuring an equal number of “No” and “Yes” observations to avoid class imbalance. To address this, we used SMOTE (Synthetic Minority Oversampling Technique), which generates synthetic observations of the minority class by interpolating between existing minority-class examples. This forces the model to learn from a more balanced distribution during training. It is important to note that SMOTE was applied only within

the training folds during cross-validation, ensuring that no synthetic data leaked into the validation or test sets.

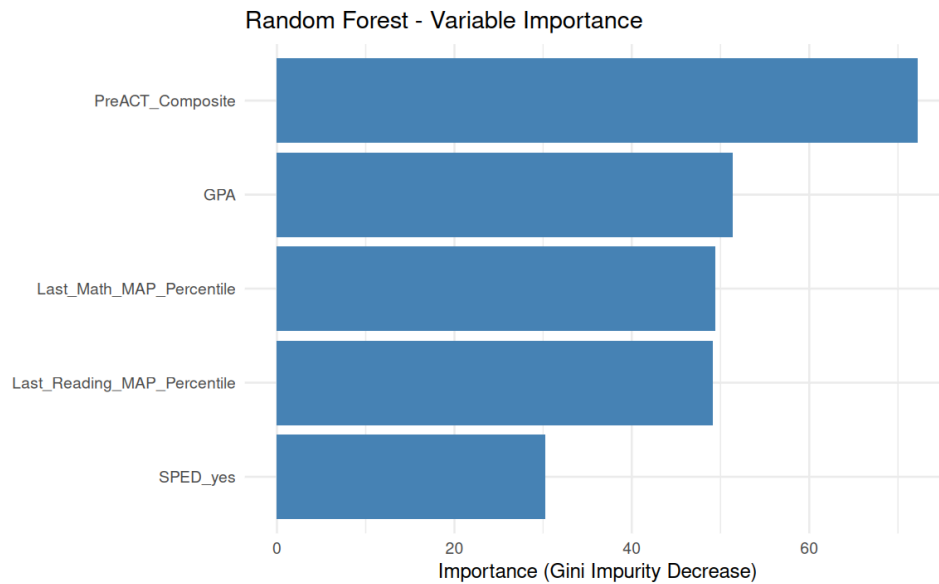


Figure 4. Feature importance within the RF model from the third run. This uses the Gini Index to determine feature importance.

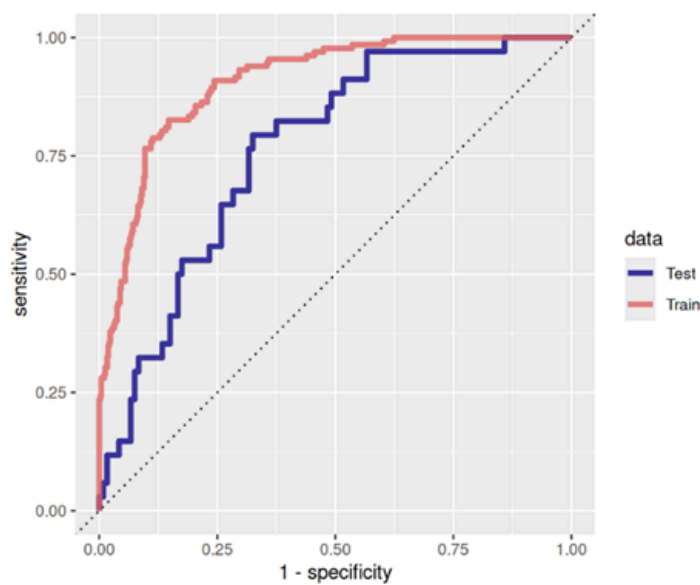


Figure 5. ROC AUC plot of the third RF run.

The table below reports four key metrics for each run. Accuracy reflects the overall share of correct predictions. Sensitivity measures how often the model correctly flags students who actually fall within the critical range; effectively, low sensitivity means that at-risk students are being missed. Specificity measures how often the model correctly identifies students outside the critical range. ROC AUC summarizes overall discrimination ability across all thresholds, where 1.0 is perfect, and 0.5 is not better than random chance.

Table of Core Performance Metrics for the Random Forest Runs						
	Second Run of Random Forest			Third Run of Random Forest		
	Train	Validation	Test	Train	Validation	Test
Accuracy	0.8914474	0.7960526	0.7922078	0.8453947	0.7680921	0.7077922
Sensitivity	0.5075758	0.1439394	0.1470588	0.8106061	0.6590909	0.5588235
Specificity	0.9978992	0.9768908	0.975	0.855042	0.7983193	0.75
ROC AUC	0.972148	0.8013114	0.7607843	0.9077063	0.8202827	0.7632353

Table of Hyperparameters for the Random Forest Runs			
	1st RF Run (Feature importance)	2nd RF Run (Hyperparameter Tuning)	3rd RF Run SMOTE
Type	Probability Estimation	Probability Estimation	Probability Estimation
Number of Trees	2000	1500	1500
Sample Size	608	608	952
Number of Independent Variables	43	13	5
Mtry	11	4	1
Target Node Size	40	40	40
Variable Importance Mode	Impurity	Impurity	Impurity
Splitrule	Gini	Gini	Gini
OOB Prediction Error	0.1388379	0.1379777	0.1259405

(Brier s.)			
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4.2 XGBOOST

XGBoost (Extreme Gradient Boosting) is another ensemble method, but unlike Random Forest, which builds trees independently in parallel, XGBoost builds trees sequentially. Essentially, it will build a tree, then create a new tree that focuses specifically on correcting the errors made by the previous ones. This often leads to higher accuracy than the Random Forest model and tends to handle imbalanced datasets more gracefully. After the Random Forest runs, XGBoost was run using the same selected features from the Random Forest feature selection. These runs used the same cross-validation setup from before in the Random Forest runs.

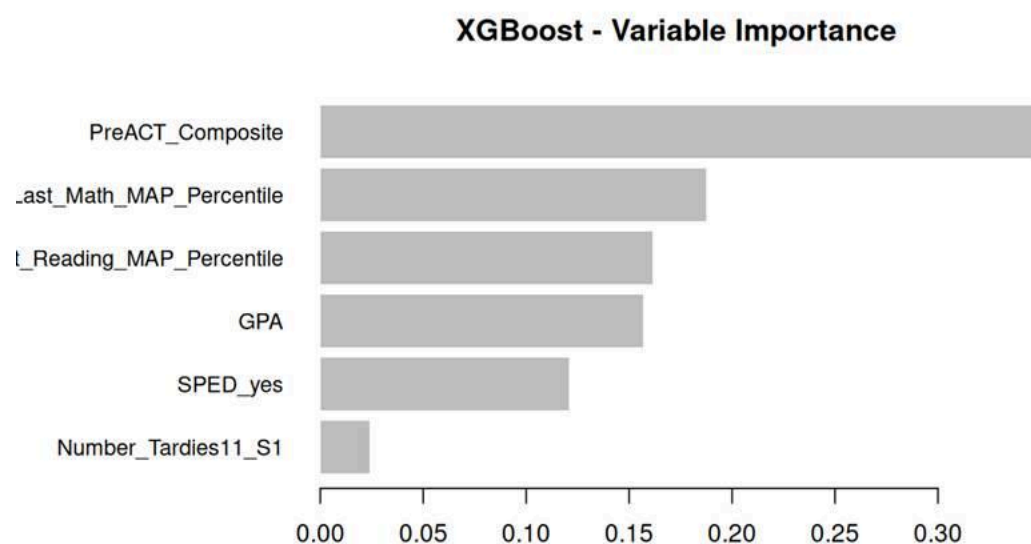


Figure 6. XGBoost first run variable importance. PreACT_Composite is more than twice the importance value of every other variable, suggesting that a student's Pre-ACT score is by far the strongest predictor of falling in the critical ACT range.

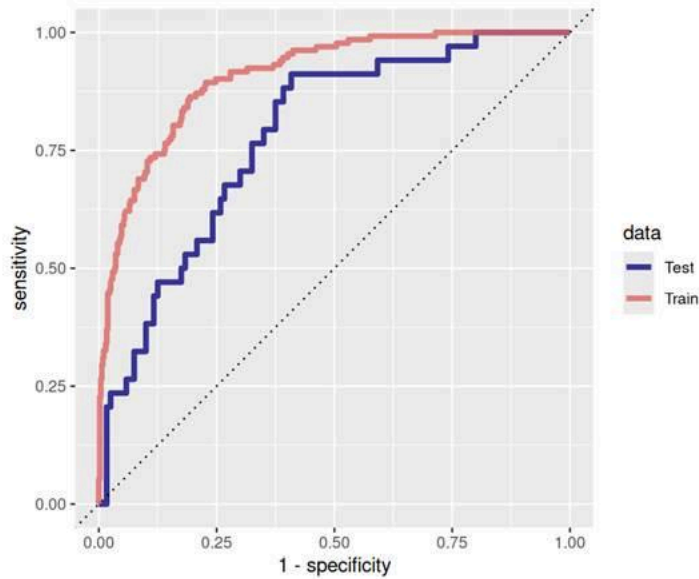


Figure 7. ROC AUC plot of the first XGBoost run.

After the first XGBoost run, a second run was conducted to address class imbalance using the same SMOTE procedure described in Section 4.1, applied only within the training folds to avoid leaking synthetic data into the validation or test sets.

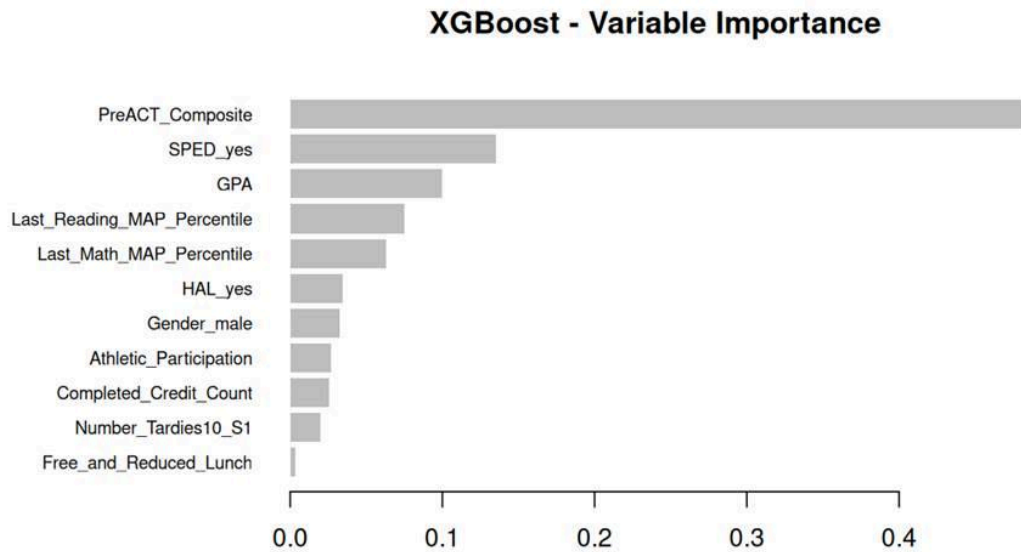


Figure 8. XGBoost second run variable importance.

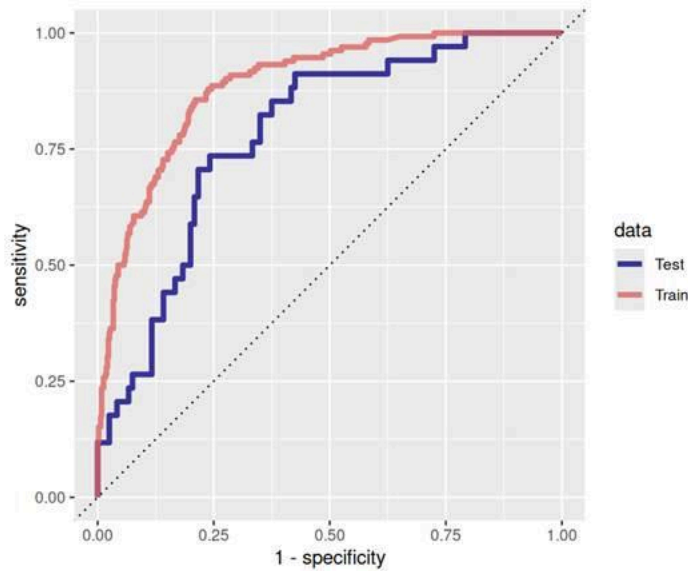


Figure 9. ROC AUC plot of the second XGBoost run.

The same four metrics reported in Section 4.1, accuracy, sensitivity, specificity, and ROC AUC, are reported below for both XGBoost runs across the training, validation, and test sets.

Table of Core Performance Metrics for the XGBoost Runs						
	XGBoost			XGBoost after SMOTE		
	Train	Validation	Test	Train	Validation	Test
Accuracy	0.8634868	0.8141447	0.7922078	0.8207237	0.7779605	0.7532468
Sensitivity	0.4924242	0.3484848	0.2647059	0.7500000	0.6287879	0.6470588
Specificity	0.9663866	0.9432773	0.9416667	0.8403361	0.8193277	0.7833333
ROC AUC	0.9081758	0.8221925	0.7803922	0.8888305	0.8157945	0.7818627

The key hyperparameters tuned were: the number of trees (how many sequential trees are built), min_n (minimum number of observations required in a terminal node), tree depth (maximum depth of each tree, controlling model complexity), and learning rate (how aggressively each new tree corrects the previous one).

Table of Hyperparameters for the XGBoost Runs		
	XGBoost	XGBoost SMOTE
Trees	330	1518
Min_n	4	11
Tree Depth	3	5
Learning Rate	0.017243933	0.002438527

4.3 PRUNED DECISION TREE

A decision tree is a model that makes predictions by learning a sequence of simple yes/no rules from the data, resulting in a branching tree structure that is easy to visualize and interpret. We used a pruned decision tree, applying a cost-complexity penalty to prevent the tree from growing too large and overfitting the training data. Unlike Random Forest and XGBoost, a single decision tree sacrifices some predictive accuracy in exchange for transparency. The decision tree was started with the feature importance from the Random Forest model. The variable importance for the decision tree confirmed the use of the selected features. The model was tuned for cost complexity using the best-fit model based on ROC AUC. Cost complexity pruning works by penalizing the tree for each additional split it makes; in effect, a higher penalty produces a smaller, simpler tree that generalizes better to new data, while a lower penalty allows a more complex tree that may overfit the data. The optimal cost complexity parameter was selected by maximizing ROC AUC across the cross-validation folds.

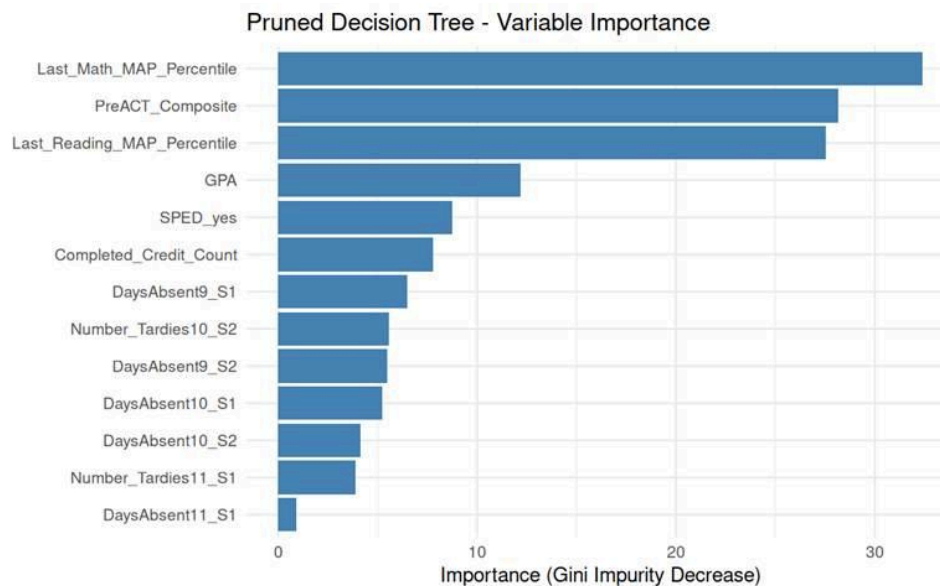


Figure 10. Pruned Decision Tree Feature Importance

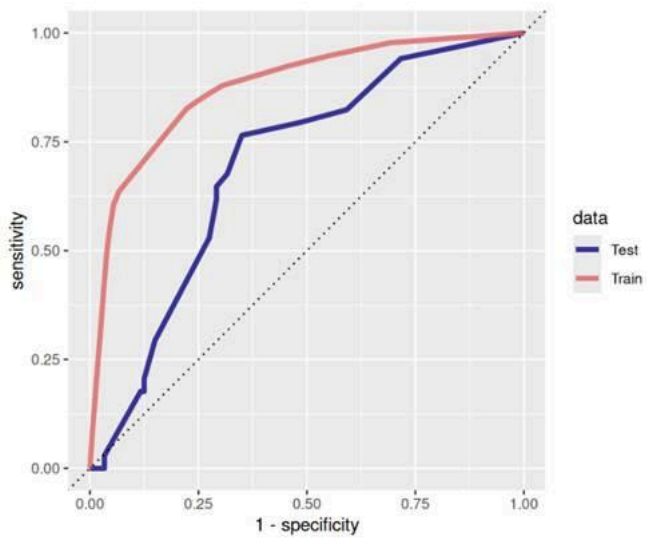


Figure 11. Pruned Decision Tree Training vs. Test ROC Curves.

Pruned Classification Tree

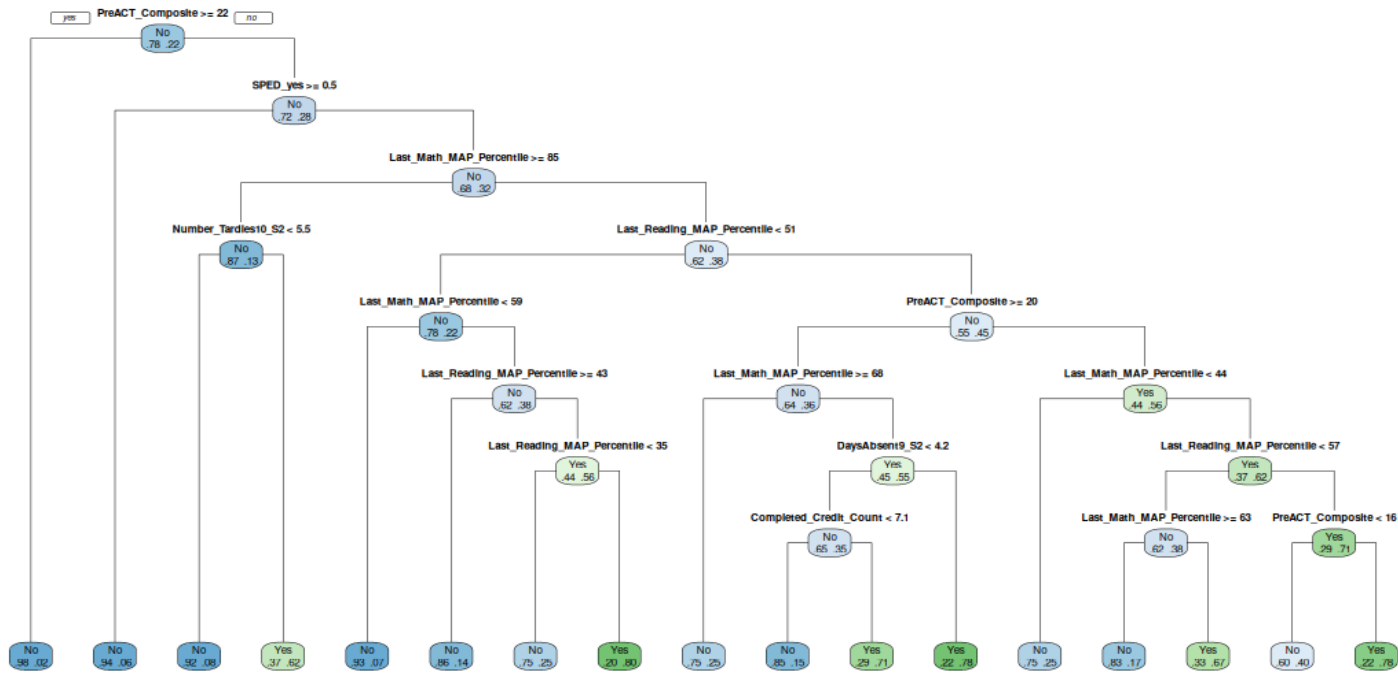


Figure 12. Pruned decision tree

The table below shows the selected cost-complexity hyperparameter, the mean cross-validated ROC AUC, and the number of folds used.

Table of Hyperparameters for the Decision Tree	
Cost Complexity	0.014923480
Mean	0.7180443
n	15

The same four metrics: accuracy, sensitivity, specificity, and ROC AUC are reported below across training, validation, and test sets.

Table of Core Performance Metrics for the Decision Tree			
	Train	Validation	Test
Accuracy	0.8717105	0.7828947	0.72727273
Sensitivity	0.6060606	0.2727273	0.20588235
Specificity	0.9453782	0.9243697	0.87500000
ROC AUC	0.8756207	0.7559763	0.69473039

Table of Core Test Performance Metrics for all the Models				
	Accuracy	Sensitivity	Specificity	ROC AUC
RF	0.7922078	0.1470588	0.975	0.7607843
RF SMOTE	0.7077922	0.5588235	0.75	0.7632353
XGBoost	0.7922078	0.2647059	0.941666	0.7803922
XGBoost SMOTE	0.7532468	0.6470588	0.7833333	0.7818627
Pruned Decision Tree	0.72727273	0.20588235	0.87500000	0.69473039

The decision tree achieved the lowest ROC AUC of all models at 0.695 on the test set, and a sensitivity of only 0.206, meaning it missed roughly 80% of the students in the critical range. However, its value lies in its interpretability. Figure 12 shows the full tree structure, which can be directly communicated to BPS staff without requiring any technical background.

Predictions by Model						
Prediction	Training Data	Random Forest	RF Smote	XGBoost	XGBoostSmote	DecisionTree
No	596	1403	806	1232	640	961
Yes	166	101	698	272	864	543
Percent-Yes	21.78%	6.72%	46.41%	18.22%	57.45%	36.1%

As stated previously, we used 5-fold cross-validation, repeated 3 times, optimizing for AUC; then we tested the optimal hyperparameter combinations. For the random forest model, there are a few different hyperparameters: mtry, mtry the number of predictors randomly sampled at each split; n_trees, the number of trees; and min_n, which is the minimum number of nodes in the terminal node before deciding on an optimal split. Each was evaluated at 5 levels, so we have 5^3 candidate hyperparameter combinations. The optimal RF hyperparameter configuration for maximizing AUC was mtry=4, trees=1500, and min_n=40, yielding an AUC of 0.7632. For XGBoost, the grid searched over similar hyperparameters. In this case, the required hyperparameters for the model are the number of trees, min_n, tree_depth, and learning rate. The best XGBoost run used 330 different trees, min_n=4, tree_depth=3, and an optimized learning rate of ~0.017. The Smote variants used the same tuning procedure, which was applied only after synthetic oversampling of the minority class in the training fold, to avoid leaking synthetic data into the validation set.

No strong distributional assumptions were imposed on the data, as both RF and XGBoost are non-parametric ensemble methods that do not assume normality of predictors or linearity between predictors and the outcome. However, several practical assumptions were made. Missing PreACT values were imputed with the mean, assuming the missingness is missing at random and that the mean is a reasonable central estimate. The synthetic imputation of balancing for the target variable procedure assumes that interpolating between existing minority-class observations yields realistic synthetic observations. Finally, the data was treated as a cross-sectional snapshot per student rather than a longitudinal time series, which assumes that the aggregated behavioral and academic variables adequately capture each student's trajectory up to their junior spring ACT.

5. MODEL INTERPRETABILITY & EXPLAINABILITY RESULTS

While the Random Forest and XGBoost models built in Section 4 are effective at classifying students, they function largely as “black box” models. They effectively produce predictions without directly explaining why a student was classified a certain way. To address this, we conducted additional explainability analyses to provide insight into what drives each classification. This is particularly important in an educational context, where school staff will need to understand not just which students are flagged, but also why, in order to take meaningful action. Two main tools were used to further clarify what is happening: waterfall plots and partial dependence plots.

5.1 RANDOM FOREST

A waterfall plot shows how each feature pushes a specific student’s predicted probability up or down from a baseline. Features shown in green increase the predicted probability of being classified in the critical range, while features shown in red decrease it. The final bar shows the model’s overall predicted probability for that student. This allows us to see exactly which factors are driving the model’s decision for any individual student, making the model’s reasoning transparent and actionable. For example, in Figure 13, Student 3372 starts from a baseline of 0.22, and their PreACT Composite score of 19, Math MAP percentile of 66, and Reading MAP Percentile of 61 all increase their probability, ultimately resulting in a predicted probability of 0.509. This is just above the classification threshold, so they result in a yes prediction. These three features were the primary drivers pushing the prediction above the classification threshold.

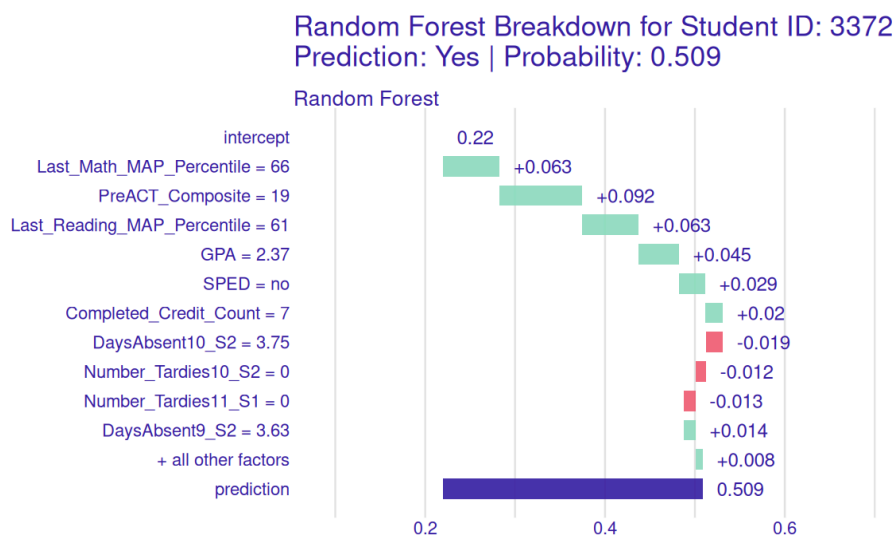


Figure 13. Waterfall plot of student 3372.

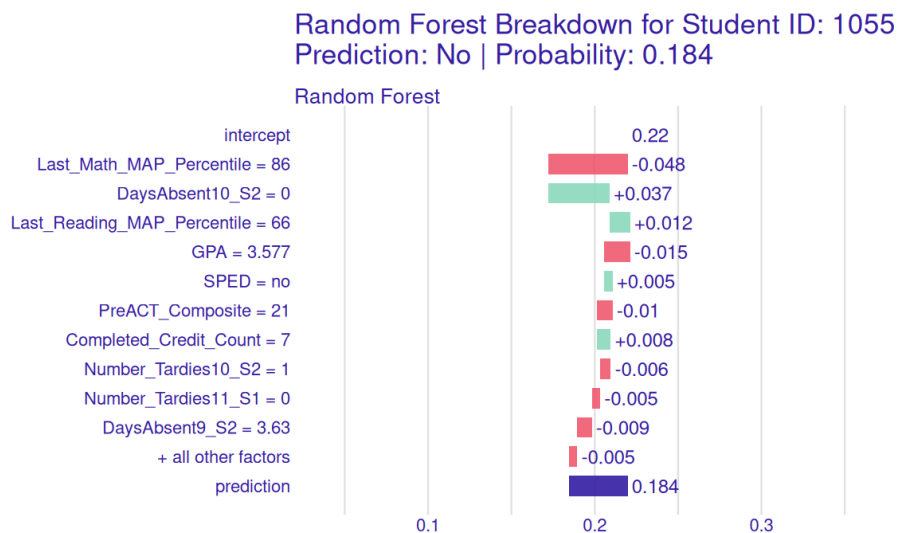


Figure 14. Waterfall plot of student 1055.

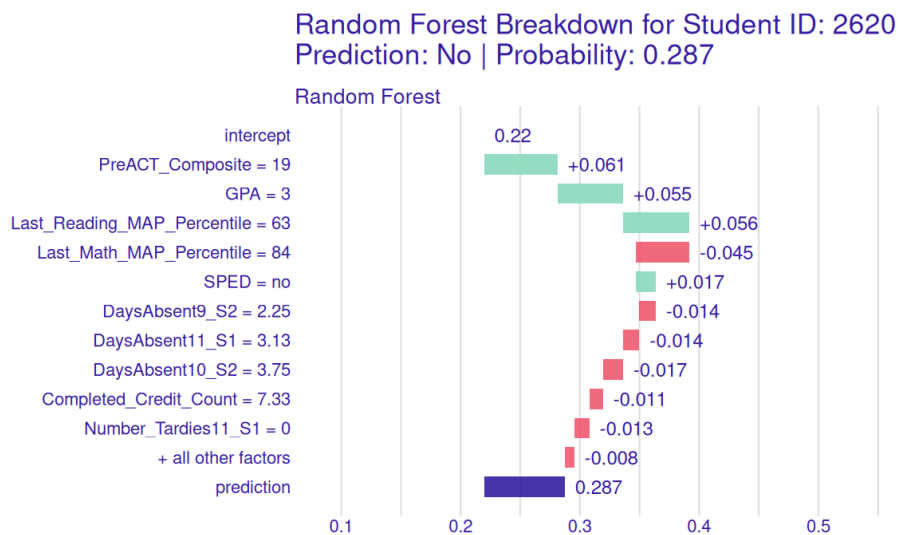


Figure 15. Waterfall plot of student 2620.

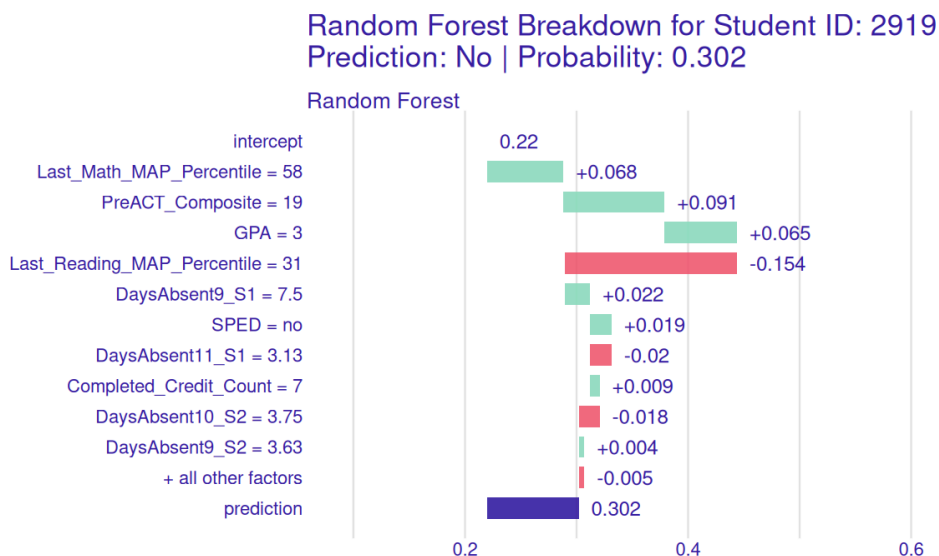


Figure 16. Waterfall plot of student 2919.

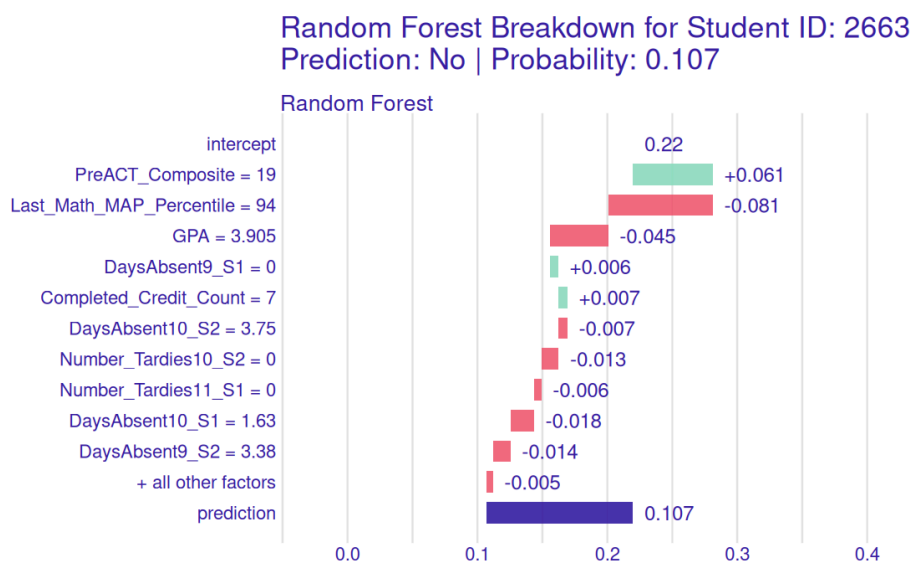


Figure 17. Waterfall plot of student 2663.

5.2 XGBoost

The same waterfall analysis was conducted for the XGBoost model. Because XGBoost was selected as the final recommended model, its individual explanations are particularly relevant for practical use by BPS staff.

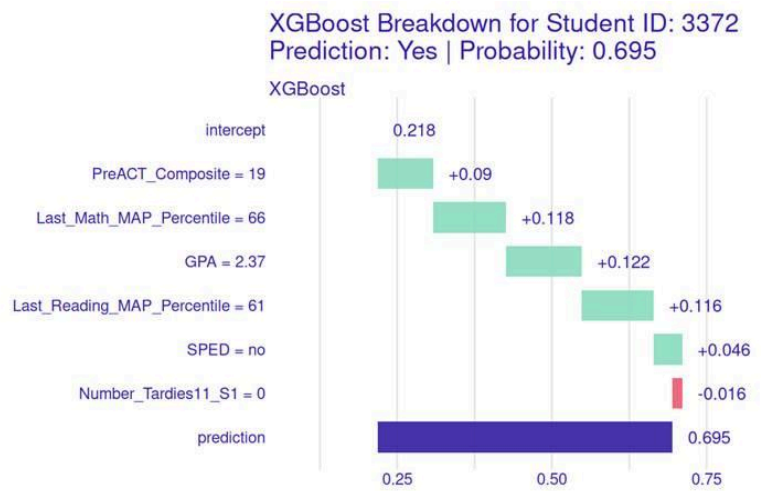


Figure 18. Waterfall plot of student 3372.

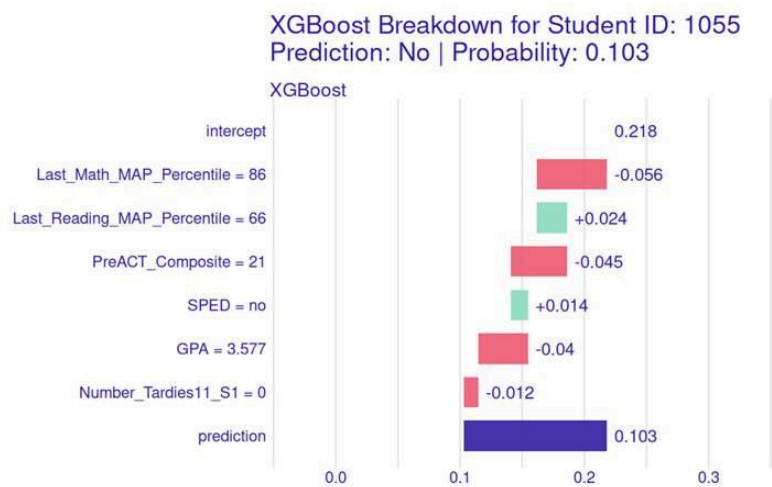


Figure 19. Waterfall plot of student 1055.

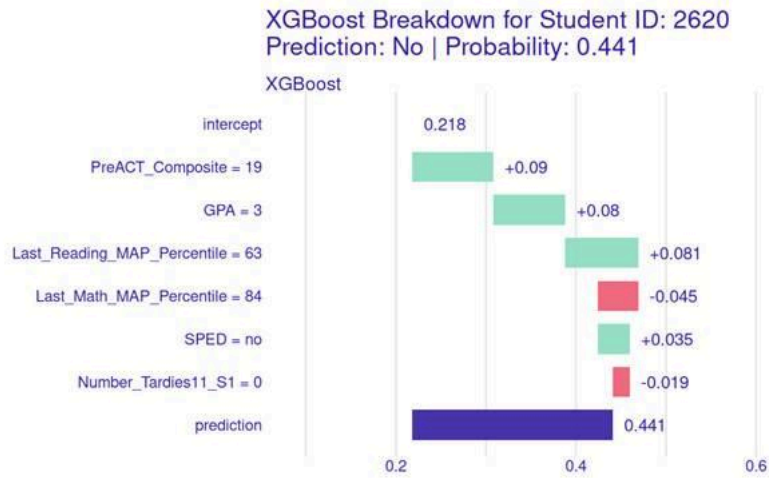


Figure 20. Waterfall plot of student 2620.

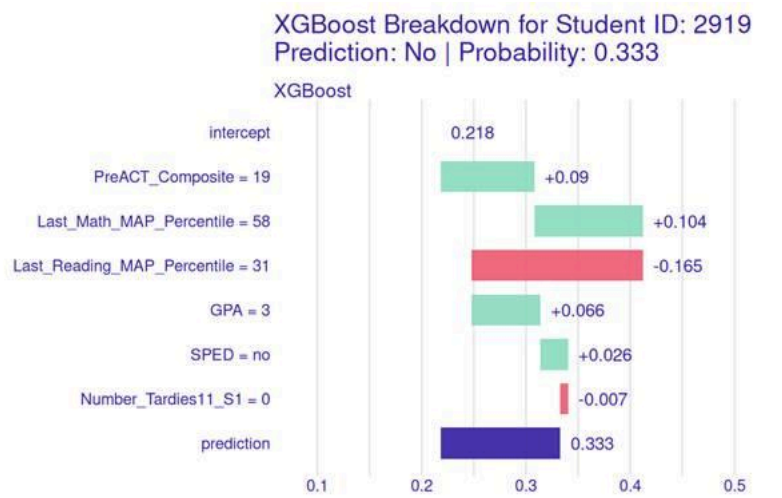


Figure 21. Waterfall plot of student 2919.

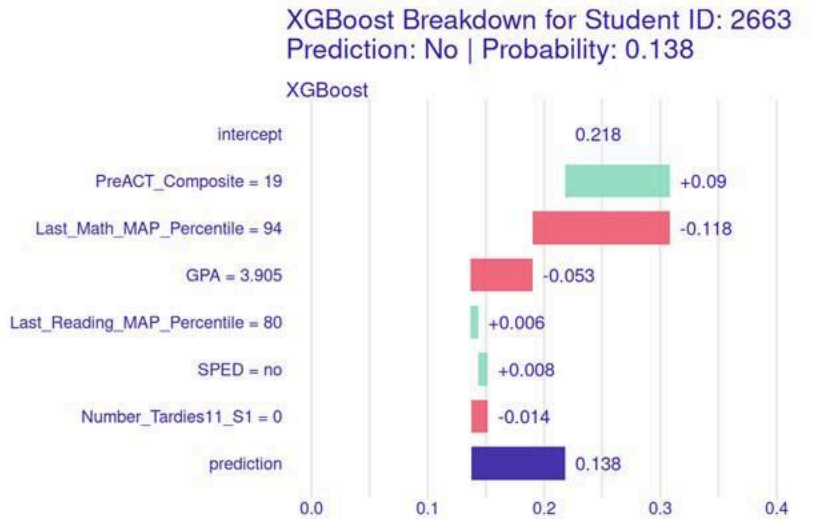


Figure 22. Waterfall plot of student 2663.

In addition to individual-level waterfall plots, we generated partial dependence plots for the XGBoost model's most important features. A partial dependence plot shows the average relationship between a single feature and the predicted probability of exceeding a critical threshold, while holding all other features constant. This allows us to understand not just which features are important, but, more specifically, how individual values relate to the outcome. For example, whether higher PreACT scores make a student more or less likely to be flagged.

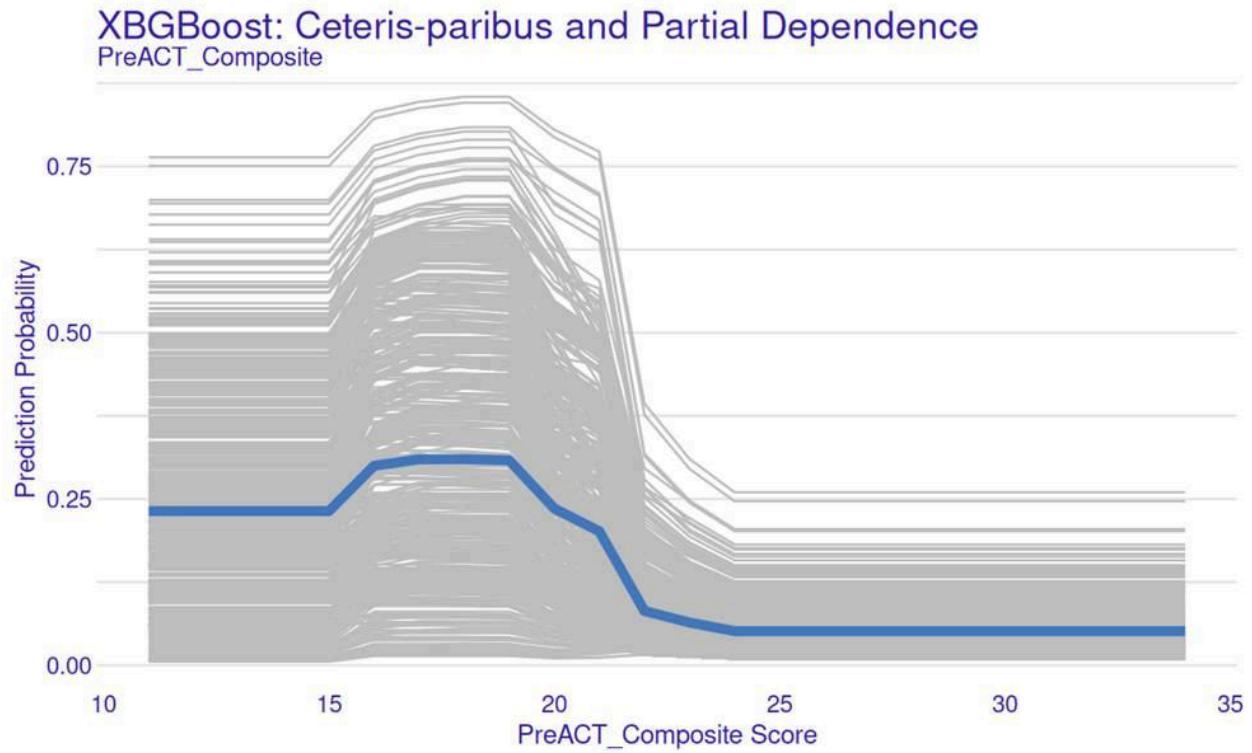


Figure 23. Partial Dependence Plot of Pre ACT Composite score feature.

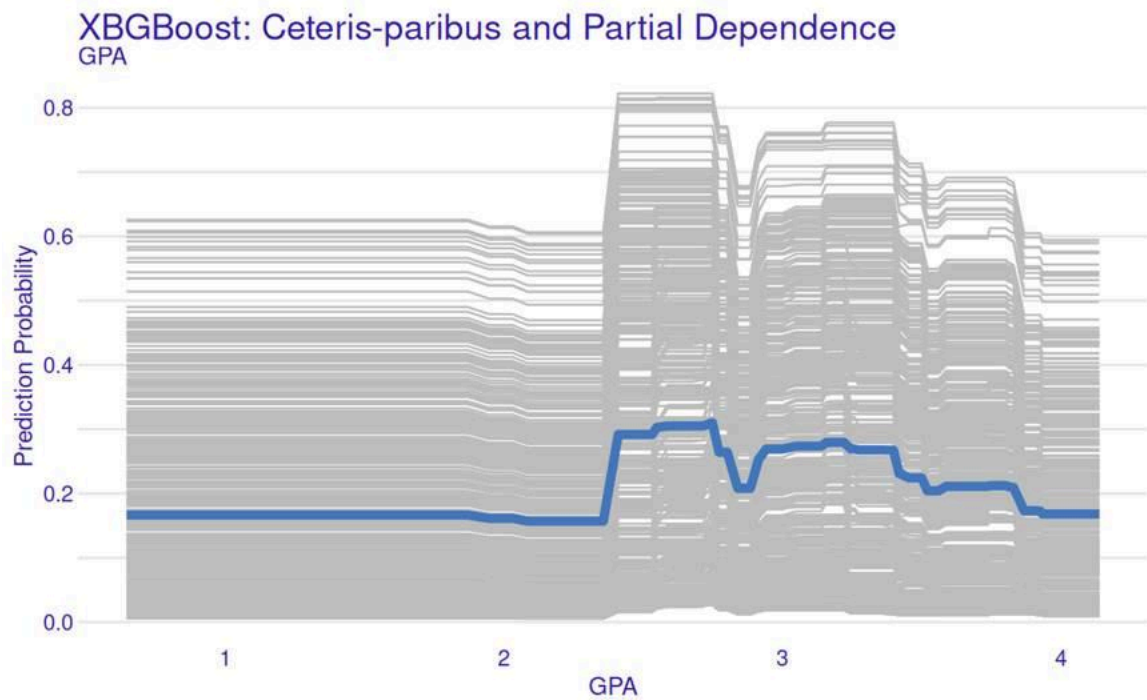


Figure 24. Partial Dependence Plot of GPA feature.

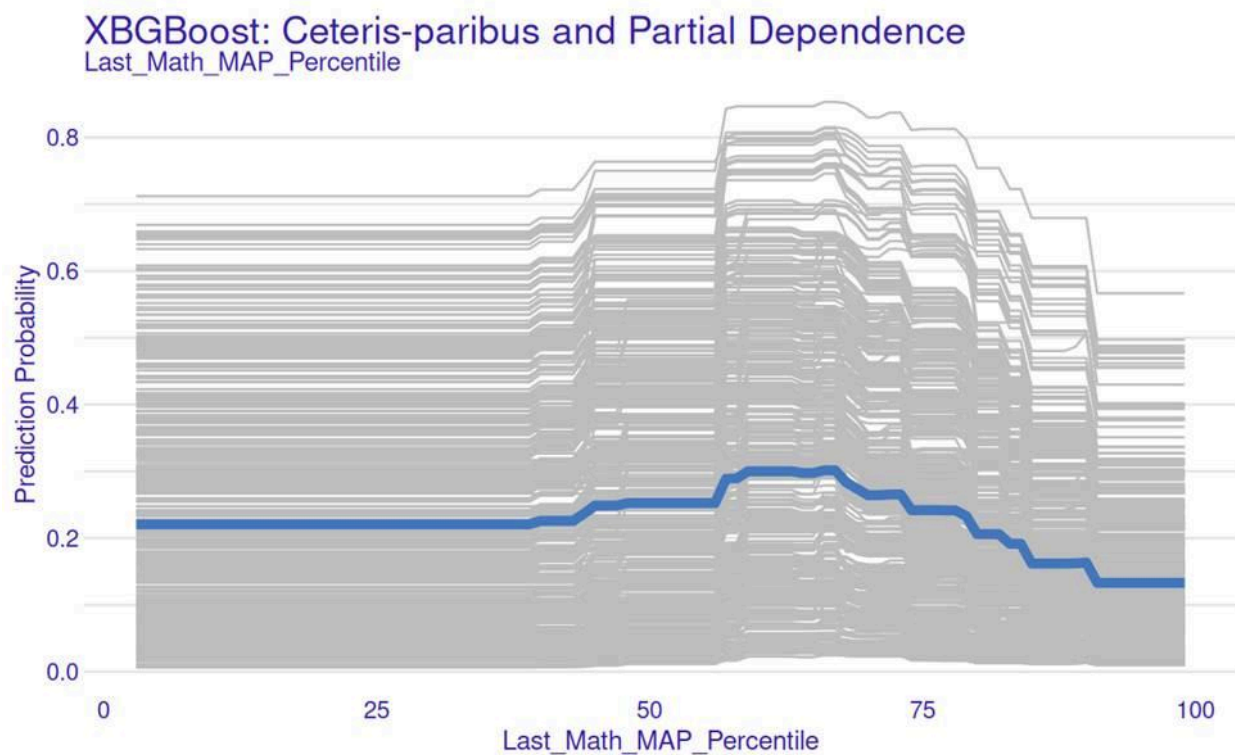


Figure 25. Partial Dependence Plot of MAP math percentile score feature.

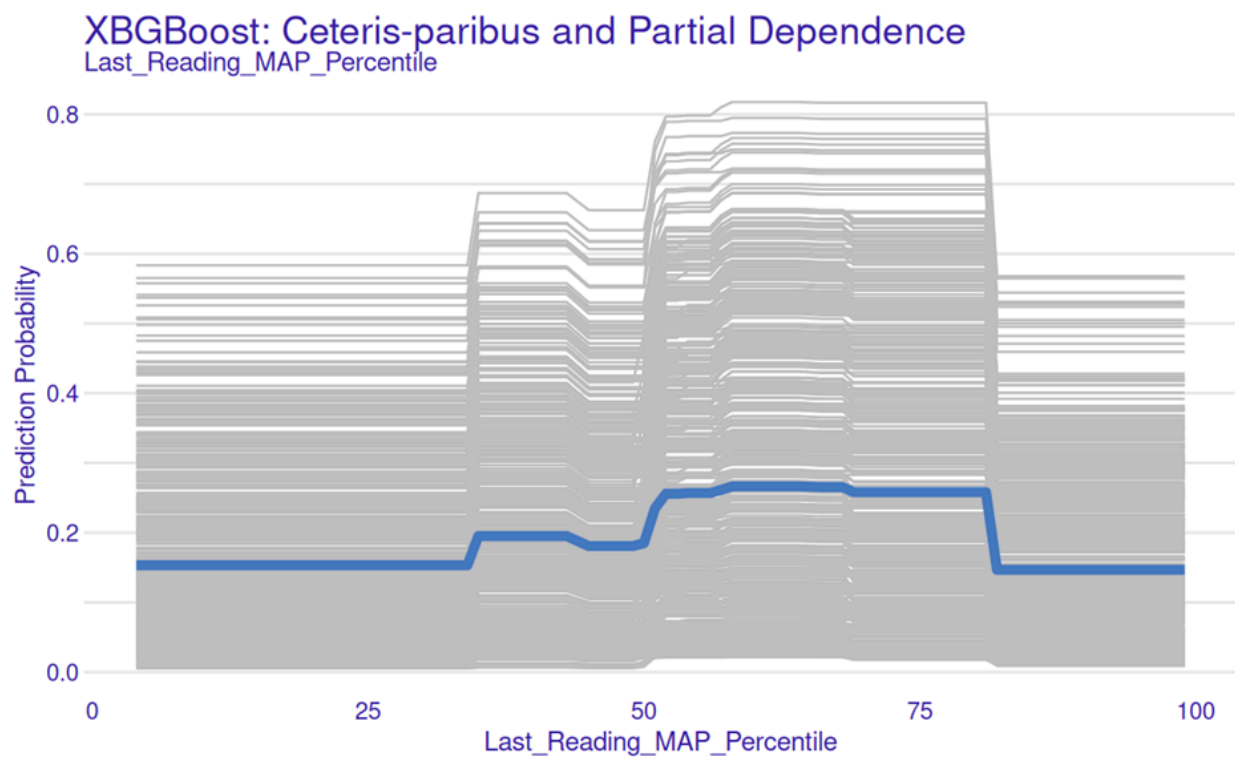


Figure 26. Partial Dependence Plot of the MAP reading percentile score feature.

5.3 DECISION TREE

Waterfall plots were also generated for the pruned decision tree. While the decision tree is already highly interpretable through its tree structure shown in Figure 12, the waterfall plots provide an additional student-level view consistent with the other models.

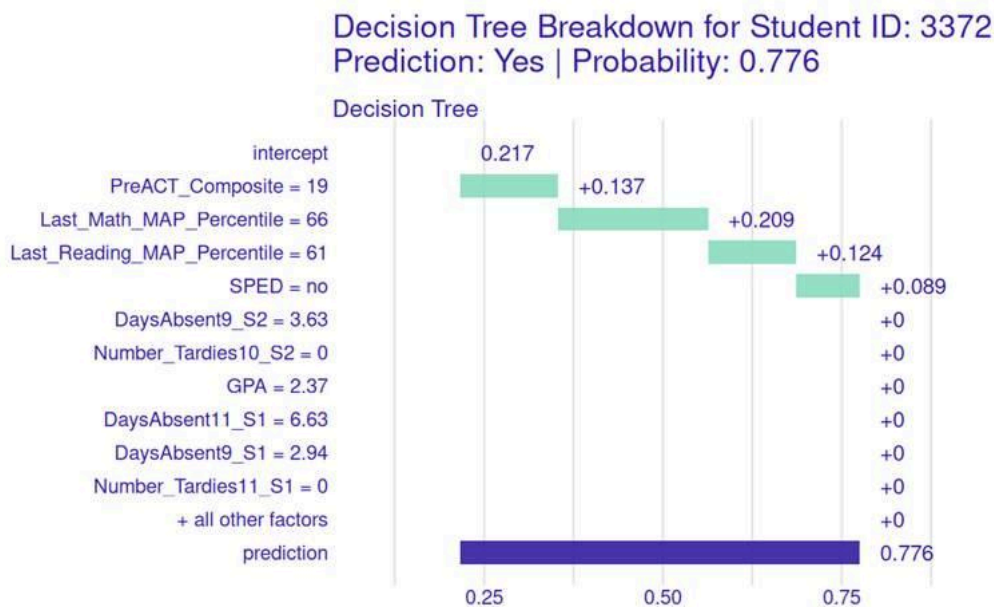


Figure 27. Waterfall plot of student 3372.

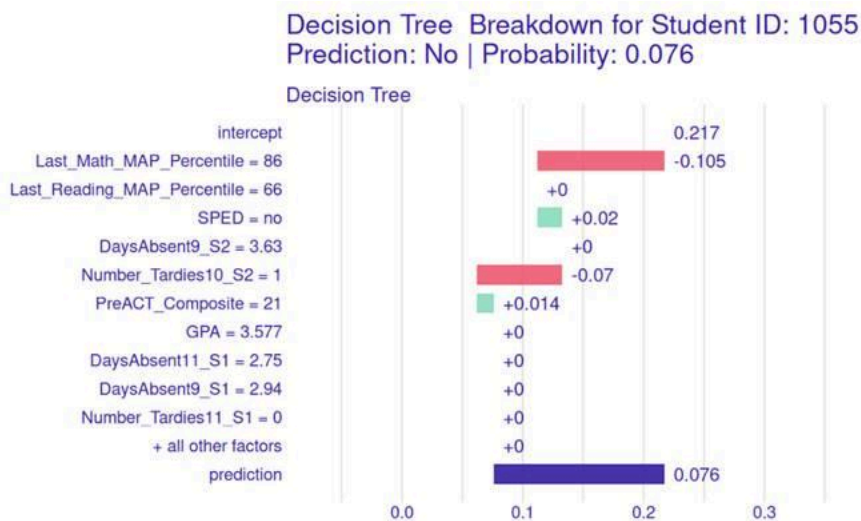


Figure 28. Waterfall plot of student 1055.

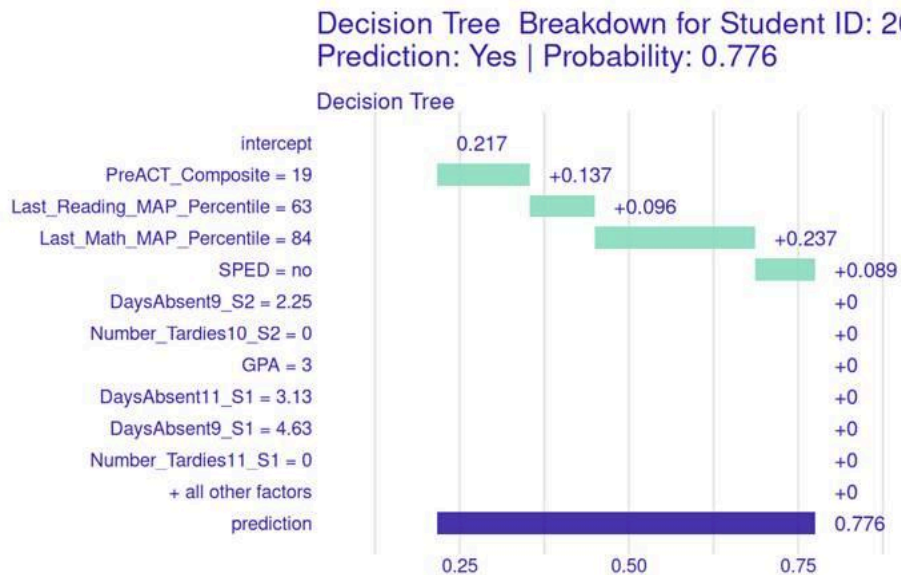


Figure 29. Waterfall plot of student 2620.

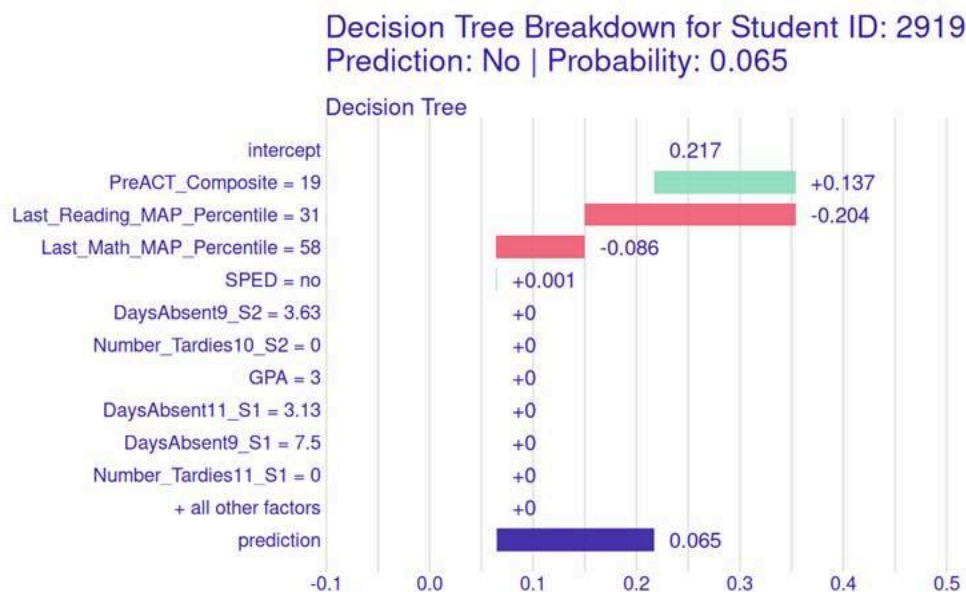


Figure 30. Waterfall plot of student 2919.

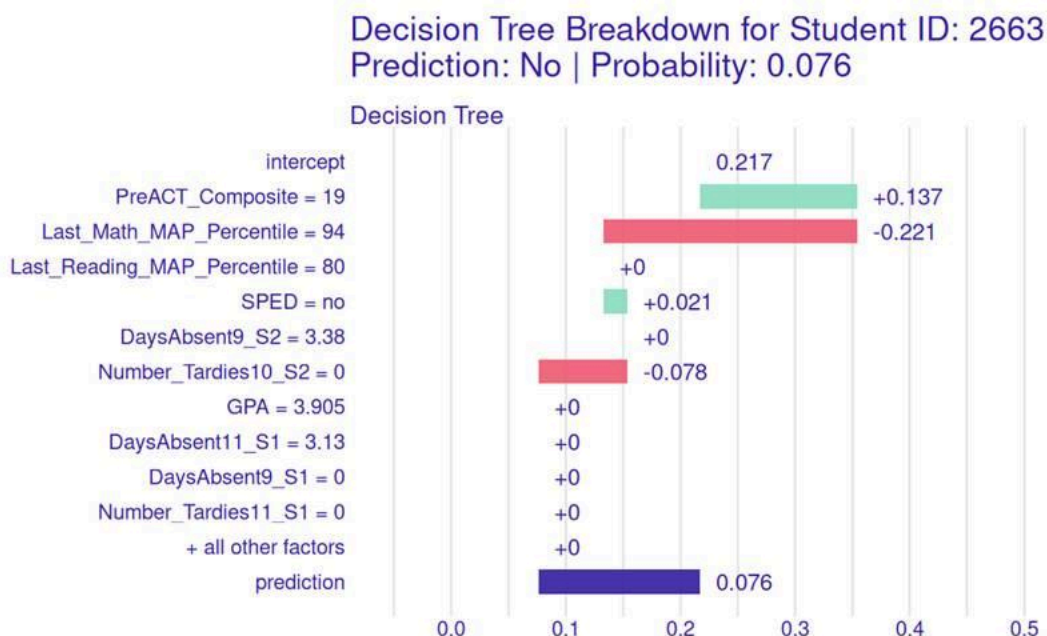


Figure 31. Waterfall plot of student 2663.

Across all three models, the waterfall plots for the five sampled students revealed consistent patterns; PreACT composite scores, GPA, and MAP percentile scores were the most frequently influential features, either strongly increasing or decreasing each student's predicted probability. The partial dependence plots for XGBoost further confirmed that PreACT scores in the 15-20 range, MAP Reading scores of 50-80, MAP Math scores of 60-75, and GPAs of 2.5-3.0 were most strongly associated with falling within the critical ACT range. These findings are discussed further in the recommendations in section 6. One particularly interesting finding from the GPA partial dependence plot is that the relationship between GPA and critical range probability is not linear. Effectively, we see the probability increase as GPA rises toward 2.5-3.0, then dips, then rebounds again at higher GPA values. This likely reflects the nature of the critical range itself: students with very high GPAs tend to score well above 20 on the ACT, placing them outside the critical range on the upper end. On the other hand, students with very low GPAs tend to score below 18, placing them at the lower end. This non-monotonic relationship suggests the critical ACT range of 18-20 captures a middle band of performance rather than a simple high or low outcome.

6. EVALUATION, RECOMMENDATION, and CONCLUSION

The primary evaluation metric used across all models was ROC AUC. This metric measures the model's ability to discriminate between the two classes across all possible classification thresholds. ROC AUC was chosen because the target variable is imbalanced, i.e., most students do not fall within the critical range of 18 to 20. Sensitivity, which is the recall for the 'Yes' class (i.e., they do fall within this range), was also closely monitored, as the cost of missing a student who is in the critical range is higher than the cost of a false positive in this educational context. A model that simply predicted 'No' for every student would achieve a 78% accuracy while being completely useless for the actual task.

Across all models, a consistent pattern emerged: models without synthetic imputation of the target variable achieved high specificity but very low sensitivity. Without class balancing, they were effective at identifying students outside the critical range but seemed to miss most students within it. This largely reflects the class imbalance; the models defaulted to predicting No, the majority class. In the RF Run 2 test set, for example, sensitivity was only 0.147, or about 15%, meaning that nearly 85% of students who actually fall in the ACT critical range would not be flagged.

Adding SMOTE substantially improved sensitivity at a modest cost to specificity and overall accuracy. The RF SMOTE model achieved a test sensitivity of 0.559, and the XGBoost SMOTE model reached 0.647, more than four times the sensitivity of the non-SMOTE RF model. The XGBoost SMOTE model also retained a respectable test ROC AUC of 0.782, indicating that it can still meaningfully rank students by risk even on held-out data. It is important to note that SMOTE was applied only within the training folds during cross-validation, meaning the test set consisted entirely of real student observations. The strong test performance, therefore, reflects genuine generalization to unseen data rather than an artifact of synthetic oversampling.

The gap between training and test performance, particularly in sensitivity, indicates some overfitting across all models. The training AUC values (~97%, ~91%, ~90%, ~88%) are considerably higher than the test AUC (~76%, ~76%, ~78%, ~78%) values across all runs. This is somewhat expected, given the relatively small sample size of 608 observations and the difficulty of the prediction task. This suggests that while the models are capable of identifying broad patterns of student risk, the high variance between training and test sets indicates that the predictors may capture some noise unique to the training sample, necessitating cautious interpretation of the exact performance ceilings.

The recommended model is XGBoost without SMOTE. While the SMOTE variant achieved higher sensitivity, it was trained on a synthetically balanced 50/50 distribution and predicted ‘Yes’ for 57.45% of students, substantially higher than the true underlying rate of approximately 22%. This suggests the SMOTE model overpredicts the minority class when applied to real-world data. XGBoost without SMOTE predicted 18.22% ‘Yes’, closely matching the true class distribution of the training data, indicating it better reflects the real-world prevalence of students in the critical range. While reduced sensitivity is a known limitation of models trained on imbalanced data, XGBoost without SMOTE achieved the highest ROC AUC among non-SMOTE models, with an AUC of 0.78, demonstrating strong overall discriminative ability without relying on synthetic data augmentation. This is the model chosen and used for submission of the final test predictions.

From the feature importance, waterfall, and partial dependence graphs, several important factors can be identified. One important finding is the predictive power of the PreACT scores. Scores that were around the critical range indicated that they were likely to be in the critical range for the actual ACT. This is best illustrated by the partial dependence plot, where PreACT scores of around 15 to 20 were more indicative of being in the critical range. Another important factor was the MAP scores in both reading and math. Based on the partial dependence plot, MAP reading scores of 50 to 80 were most likely to indicate that a student would fall in the critical range. For the MAP math, scores of 60 to 75 were most likely to indicate a student falling in the critical range. For a student’s GPA, a score of about 2.5 to 3.0 was most likely to indicate whether a student is to fall in the critical range. In addition to these features, the models also identified tardies and absences as important. These features were used in various ways to determine

whether a student was in the critical range, and can be seen to add, subtract, or be neutral in a model's classification. This is likely due to the specific range selected for ACT's critical range, as the full ACT range extends both above and below the critical range. Another important feature was SPED, as indicated by the feature importance and waterfall graphs. This feature, indicating 'no', usually increases the probability that a student will fall within the critical range. From this evidence, several recommendations can be made. Students with PreACT scores of 15 to 20 are most likely to improve their ACT scores and should be given resources to succeed on the standardized test. MAP scores also indicate students who are most actionable: MAP reading scores of 50 to 80 and math scores of 60 to 75. GPAs can also identify students with actionable reasons to improve their ACT scores, namely, GPAs of 2.5 to 3.0. Additionally, reducing tardies and absences may be valuable towards a student's success on the ACT.

This project explores the classification of students towards being in the critical range for their ACT scores. This ACT range of 18 to 20 indicates a successful ACT for a student. Here, we built, trained, and tested models to classify students into this critical range. These models included random forests, XGBoost, and decision trees. These models performed well toward their objective, and an XGBoost model was selected as the best model for further testing. In addition to the model generation, the models were used to explain why students may fall within the critical range. Several important factors, including their PreACT scores, GPA, MAP standardized test scores, tardies, and absences, were identified as being the most important. These models and their explanations expand understanding of students and their ACT scores, hopefully improving their outcomes.

7. REFERENCES

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